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March 23, 2026

Maki People – Shiro Bias Audit Report



Holistic AI





REPORT ON MAKI PEOPLE'S SHIRO AND ON THE SUITABILITY OF ITS DESIGN AND OPERATING EFFECTIVENESS RELEVANT TO AUTOMATED EMPLOYMENT DECISION TOOL BIAS AND ASSOCIATED RISKS

Pursuant to Reporting on the New York City Local Law 2021/144 – Title: A Local Law to amend the administrative code of the city of New York, in relation to automated employment decision tools.

March 23, 2026



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Assertion of Maki People Management

Purpose

We have commissioned Holistic AI Inc (“Service Auditor”) to conduct an independent and impartial audit (“Bias Audit”) of Maki People’s (“Service Organization”) Shiro under New York City Local Law 144 - A Local Law to amend the administrative code of the city of New York, in relation to automated employment decision tools (dubbed here as “NYC Bias Audit” or “Legislation”).

Scope

We have shared with the Service Auditor the appropriate documentation, code, and data necessary to conduct the audit.

- **Shiro** — Shiro is an AI-assisted decision-support assessment platform that evaluates 19 soft skills using two standardized open-ended questions per skill (broad and scenario-based), applying light preprocessing, Behaviourally Anchored Rating Scales, and a large language model to assign dimension-level scores which are aggregated into evidence-based scorecards with quantitative results and qualitative explanations to support human hiring decisions, and it does not autonomously make hiring decisions.

We confirm, to the best of our knowledge and belief, there is no relevant audit information in our possession, custody, or control that we did not provide to the Service Auditor and reflects how Shiro operated from January 1, 2026 to March 23, 2026.

Aiden Loe

Full Name

Head of Science

Title

Cambridge

27/03/2026

Location and Date

Aiden Loe

[Aiden Loe \(Mar 27, 2026 10:07:51 GMT\)](#)

Signature



Executive Summary

Purpose

Holistic AI Inc (“Service Auditor”) have been commissioned to conduct an independent and impartial audit (“Bias Audit”) of Maki People’s (“Service Organization”) Shiro under [NYC Local Law 144](#) - A Local Law to amend the administrative code of the city of New York, in relation to automated employment decision tools (dubbed here as “NYC Bias Audit” or “Legislation”). We have also been requested to review the Service Organization’s internal controls with respect to Adverse Impact, Controls and Transparency. In this sense, we want not only to check the minimum required by the legislation but also to improve how our organization operates and builds technology. The outputs of this Bias Audit, in accordance with the Legislation are:

- Bias Audit report (this document).
- Summary of Results, shared through a web application and to be displayed in their website or other forums.
- If needed, recommendations on internal controls.

Scope

In accordance with the Legislation, the Service Organization has conducted an internal assessment and provided Holistic AI with the following “System” for the Audit:

- **Shiro** — Shiro is an AI-assisted decision-support assessment platform that evaluates 19 soft skills using two standardized open-ended questions per skill (broad and scenario-based), applying light preprocessing, Behaviourally Anchored Rating Scales, and a large language model to assign dimension-level scores which are aggregated into evidence-based scorecards with quantitative results and qualitative explanations to support human hiring decisions, and it does not autonomously make hiring decisions.

Key Findings

The bias audit evaluated scoring outcomes generated by the Shiro - NYC Bias Audit Workflow system across gender and race/ethnicity groups in accordance with the methodology outlined under New York City Local Law 144 and DCWP guidance. Standalone analysis showed scoring rates of 0.50 for female individuals and 0.50 for male individuals, corresponding to an impact ratio of 0.99. Across race and ethnicity groups, impact ratios ranged between 0.85 for White individuals to 1.00 for American Indian or Alaska Native individuals. Intersectional analysis combining gender and race/ethnicity groups produced impact ratios ranging from 0.82 for male White individuals to 1.00 for female American Indian or Alaska Native individuals. Notably, some impact ratios for White individuals and their intersectional groups approached but did not fall below the 0.80 four-fifths threshold. Overall, all observed impact ratios met the reference threshold, indicating consistent scoring outcomes across the demographic groups evaluated within the dataset used for this audit.

Recommendations

Annual audit as required by NYC LL144 with the next date set at March 23, 2027.

Although no impact ratios fell below the 0.80 threshold indicating adverse impact, some groups such as White individuals and related intersectional subgroups approached this boundary, suggesting areas for continued vigilance. The company should maintain robust internal monitoring processes to routinely assess scoring outcomes across all protected groups to detect any emergent disparities promptly. Implement periodic fairness reviews integrated into model performance evaluations to ensure ongoing compliance with NYC Local Law 144. It is important to document all monitoring



activities and remediation efforts thoroughly to meet regulatory expectations. Additionally, consider implementing bias mitigation strategies proactively, such as algorithmic adjustments or fairness-aware training techniques, especially targeting subpopulations with impact ratios nearing concern levels. Engage diverse stakeholders in reviewing outcomes and refining decision criteria to uphold equitable treatment. By establishing a continuous fairness governance framework and committing to transparent reporting, the organization can sustain equitable results over time and reduce risk of inadvertent bias in future deployments.



1. Auditing Methodology

The Bias Audit of Maki People's system (see Table 1 and the Appendix report for definitions) conducted by Holistic AI as part of the requirements of New York City Local Law 144 is comprised of five major stages (Figure 1):

1. **Onboarding** – the main requirements of the audit, system qualification, and available data are discussed with the Service Organization.
2. **Information and data sharing** – relevant materials such as system and company descriptions, technical manuals, and data are shared with Holistic AI.
3. **Data cleaning and analysis** – Data that is received by Holistic AI is cleaned where needed and analyzed using the appropriate metrics for the type of data provided.
4. **Report** – Holistic AI curates a report on the audited system that includes system description, impact ratios, and recommendations.
5. **Ongoing monitoring** – Holistic AI issues a web application that can be used to access and share the *Summary of Results* with interested parties. It also can be used to periodically monitor the system and re-audit it annually or after major changes.

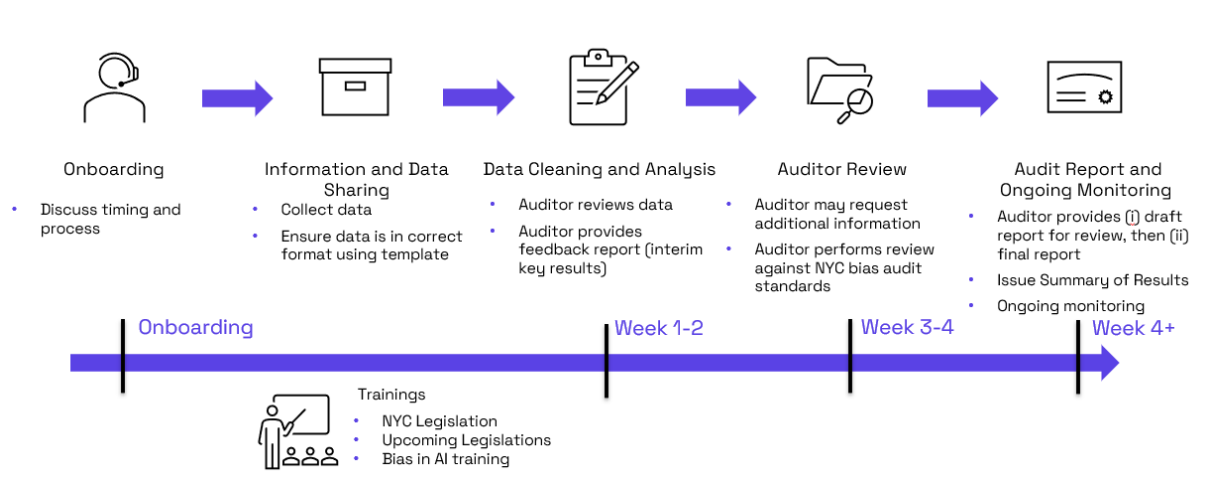


Figure 1. Typical Bias Audit process timeline and example of activities conducted

Additionally, trainings regarding NYC Bias Audit Law, Bias in AI, and upcoming legislations are provided to the Service Organization developers and management. An optional step involves reviewing the internal controls implemented by the Service Organization regarding Adverse Impact, Bias Mitigation and Transparency. The main aspects of the key controls verified are:

- **Preventative Controls.** Documentation, resources, and internal policies.
- **Detective Controls.** Metrics and monitoring technology.
- **Corrective Controls.** Internal tools and other safeguards.

The following sections unpack each of these five stages.

1.1. Onboarding

During the onboarding phase the Service Organization and Holistic AI discuss the requirements of the audit as well as identify the key systems and extract the associated data for conducting the audit. This includes reviewing the definitions of an Automated Employment Decision Tool (AEDT) under New York City Local Law 144. See Table 1 for the definitions used to determine qualifications.

**Table 1. Definitions**

Term	Definition	Source
Automated Employment Decision Tool (AEDT)	Any computational process, derived from machine learning, statistical modelling, data analytics, or artificial intelligence, that issues simplified output, including a score, classification, or recommendation, that is used to substantially assist or replace discretionary decision making for making employment decisions that impact natural persons.	Local Law 144
To substantially assist or replace discretionary decision making	(i) to rely solely on a simplified output (score, tag, classification, ranking, etc.), with no other factors considered; (ii) to use a simplified output as one of a set of criteria where the simplified output is weighted more than any other criterion in the set; or (iii) to use a simplified output to overrule conclusions derived from other factors including human decision-making.	DCWP adopted rules
Machine learning, statistical modelling, data analytics, or artificial intelligence	A group of mathematical, computer-based techniques: i) that generate a prediction, meaning an expected outcome for an observation, such as an assessment of a candidate's fit or likelihood of success, or that generate a classification, meaning an assignment of an observation to a group, such as categorizations based on skill sets or aptitude; and ii) for which a computer at least in part identifies the inputs, the relative importance placed on those inputs, and, if applicable, other parameters for the models in order to improve the accuracy of the prediction or classification.	DCWP adopted rules
Employment decision	To screen candidates for employment or employees for promotion within the city.	Local Law 144
Employment agency	All persons who, for a fee, render vocational guidance or counselling services.	6 RCNY § 5-249.

The legislation states that audits should include, at a minimum, testing for disparate impact against component 1 categories required to be reported by employers under subsection (c) of [section 2000e-8](#) of title 42 of the United States Code as specified in [part 1602.7](#) of title 29 of the code of federal regulations. [Rules adopted](#) by the New York City Department of Consumer and Worker Protection further clarify this, explicating that the audits, at minimum, should cover each race/ethnicity and sex category that [is required to be reported](#) to the Equal Employment Opportunity Commission (EEOC), with the updated and adopted versions of the rules also specifying that intersectional analysis must be carried out in addition to standalone analysis. More information about each component 1 category is displayed in Table 2.

Table 2. Component 1 categories required to be reported by employers under subsection (c) of section 20-871

Component 1 Category	Subcomponents	Description
Sex/Gender	Male	This category includes individuals who identify as male.
	Female	This category includes individuals who identify as female.
	Other	This is an optional category. When included, it reflects individuals who identify as neither male nor female,



		including non-binary and other gender identities.
Race/Ethnicity	Hispanic or Latino	This category includes individuals of Cuban, Mexican, Puerto Rican, South or Central American, or other Spanish culture or origin, regardless of their race.
	White	This category includes people with origins from Europe, the Middle East, or North Africa
	Black (or African American)	This category includes individuals that have origins in any of the black racial groups of Africa.
	Native Hawaiian or Other Pacific Islander	This category includes anyone with origins in Hawaii, Guam, Samoa, or other Pacific Island.
	Asian	This category includes individuals with origins in the Far East, Southeast Asia, or the Indian Subcontinent. This includes Cambodia, China, India, Japan, Korea, Malaysia, Pakistan, the Philippine Islands, Thailand, and Vietnam.
	American Indian or Alaska Native	This category includes people with origins in any of the original peoples of North and South America (including Central America) who maintain tribal affiliation or community attachment.
	Two or more races	This category includes anyone who identifies with more than one race. This does not include those identifying as Hispanic or Latino.

If available in the dataset and depending on the context, impact ratios can also be estimated for age (under 40 and 40 and over), disability status, veteran status, and others on a case-by-case basis.

1.2. System information and data sharing

During the system information and data sharing phase the Service Auditor aims to methodologically understand the system, the key information it uses, and the outcome produced. The correct classification of the system type will guide the data and information shared between the Service Organization and the Service Auditor. On that basis, and following the DCWP guidance, we can broadly separate the systems or processes based on the outcome generated as per Table 3.

Table 3. System type, examples, and mathematical/process characterization

System Type	Example	Mathematical Formula or Process
Selection	Pass/Reject or Promoted/Not Promoted	In mathematical form one can express the outcome $y_i \in \{0,1\}$ of the whole process as a composition of functions: $y_i = (f \circ g \circ \dots \circ h)(\mathbf{x}_i)$ for a certain individual i with key information $\mathbf{x}_i$. If y_i is equal to 1, it indicates that the individual was successful, and vice-versa 0. Each function



		$f, g, \dots, h$ represents a system or human decision making process as part of the selection/screening process.
Binary outcome	Recommendation or tagging of candidates or employees, such as high/low performer, approved/rejected, etc.	The system producing the outcome $y_i \in \{0,1\}$ can be expressed as $y_i = f(\mathbf{x}_i)$ for a certain individual i with key information $\mathbf{x}_i$. If y_i is equal to 1, it indicates that the individual was successful, and vice-versa 0.
Continuous outcome	Scoring of candidates or employees, such as skills, behavior assessments, candidate matching, etc.	The system producing the outcome $y_i \in [a, b]$ can be expressed as $y_i = g(\mathbf{x}_i)$ for a certain individual i with key information $\mathbf{x}_i$. The closer y_i is to b , it indicates that the individual is more likely to be successful, and vice-versa if near to a .

Following the adopted Rules, one needs to report a description of the system, key information, and collect adequate data for impact ratio estimation. Table 4 displays the key elements collected.

Table 4. System information and data sharing

Item	Reason	Typical Source	Key Elements
System Description	To describe the System used from a user and technical perspective	- Website - User manual - Terms of Service - Technical documentation - Interview with product and business teams	General information about the System processing, the characteristics used for outcome generation, the business case or use
Key Information	To describe the information used by the System or other process	- Technical documentation - Interview with data science, IOs, or engineering teams - API or other infrastructure document or diagram	Variables that are being used by the System to provide an outcome for a given individual
Historical Data	To calculate the System or other process impact ratios	- Internal database - HR Information System - Supplier database	- Individual Identifier - Job Title / Role Identifier - Location - Gender - Race/Ethnicity - Score (Optional) - Company Identifier - Age - Disability Status - Veteran Status

To ensure these items are in the correct format, a template is provided by Holistic AI for the Service Organization to fill in. When historical data is shared in an anonymized fashion, we request the Service Organization spot-checks to validate that the data is historical and not test or synthetic information. Information is also collected on how the company operates and the contexts that the system is used. Some of this information may be sourced from the Service Organization's website.

In accordance with DCWP Rules, in the absence or if the historical is insufficient to conduct a statistically significant bias audit, Holistic AI will work with the Service Organization to use adequate Test Data.



This phase can also include the optional internal controls review to establish whether the Service Organization has taken any steps to mitigate risks surrounding bias and transparency. To do this, a questionnaire curated by Holistic AI is shared with the Service Organization for completion. Responses are then reviewed by Holistic AI's auditing team to determine the level of action taken by the Service Organization and inform recommendations on how to strengthen or maintain controls.

1.3. Data quality assurance and impact ratio estimation

Data quality assurance

Once the data from the Service Organization has been received, Holistic AI investigate inconsistencies and other issues. This includes, for example, ensuring that all labels for protected characteristics are consistent (e.g., ensuring uniform coding of the protected attributes) and assessing whether the column labels are congruent with the column contents (e.g., ensuring that nationality is not present in the ethnicity column).

Missing data

Under the adopted version of the rules, bias audits must indicate the number of individuals the System assessed that are not included in the required calculations because they fall within an unknown category. In other words, if datapoints are removed from the dataset due to missing gender/ethnicity information, the number of individuals with missing data that are not analyzed during the audit must be specified in the report and summary of results.

Impact ratio calculation

After the data is cleaned and assured, impact ratios are then calculated, using DCWP Rules guidance. The rate calculation, depending on the outcome of the system and data analyzed, is done per Table 5.

Table 5. Rate calculation per system type

System Type	Rate Calculation
Selection or any binary outcome	Given a component category C and subcomponents/groups k, one can estimate the selection/scoring rate SR_{C_k} by: $SR_{C_k} = \frac{\sum_{i \in C_k} y_i}{n_{C_k}}$
Continuous outcome	First, calculate the median med across all the samples, regardless of the group that the individual i belongs: $med = median(y_1, \dots, y_i, \dots, y_n)$. Then, estimate the scoring rate using the following formula: $SR_{C_k} = \frac{\sum_{i \in C_k} 1(y_i > med)}{n_{C_k}}$ where $1(y_i > med)$ is an indicator function, where it returns 1 if $y_i > med$, otherwise 0.

Impact ratios are then estimated by using each SR_{C_k} using the formula:

$$IR_{C_k} = \frac{SR_{C_k}}{\argmax_{k \in C} SR_{C_k}}$$

These metrics are used for both standalone/category-level analyses based on individual subgroups (e.g., male vs female) and intersectional analysis based on the intersection of subgroups (e.g., black



male vs black female), in line with the rules put forward by New York City's Department for Consumer and Worker Protection (DCWP).

Impact ratios can be calculated at different aggregation levels, depending on the circumstances of the data and requirement by the Service Organization. Table 6 displays the different levels, rationale and adjustments that may be required for impact ratio estimation.

Table 6. Aggregation levels, rationale, and impact ratio estimation

Aggregation Level	Rationale	Impact Ratio Calculation
System-level (cross-employers/employment agency audit)	Used by System providers when if such employer or employment agency provided historical data from its own use of the System to the independent auditor conducting the bias audit or if such employer or employment agency has never used the System	Requires adequate control and segmentation for job-level, company-level, industry, location, and many other factors in the results
Company-level (employer or employee agency)	Used by employers and employee agencies when the System usage per job/role is insufficient for a statistically significant audit	Requires adequate control for job-level results
Job/Role-level	Used by employers and employee agencies when the System usage per job/role is sufficient for a statistically significant audit	Sample size can be a challenge

For system and company aggregation levels, controlling and adequate outlier management and data segmentation may be required to produce impact ratios estimates that are more statistically efficient, that is, with less variance. Therefore, given a set of controlling variables $\mathbf{d}_i$ and a controlling model g , one can produced the adjusted outcome y_i^* for aggregation by using the formula:

$$y_i^* = h(y_i, g(\mathbf{d}_i))$$

Typical $\mathbf{d}$ include representing job/roles as dummy variables. Table 7 displays how controlling is performed for a given set of controlling variables and requirements.

Table 7. Controlling type, formula and hyperparameters

Norming type	Formula	Hyperparameters
Standard scaling	$y_i^* = \frac{y_i - \mu_{\mathbf{d}}}{\sigma_{\mathbf{d}}}$	$\mu_{\mathbf{d}}$: average of y_i for a given controlling variable $\mathbf{d}$ $\sigma_{\mathbf{d}}$: standard deviation of y_i for a given controlling variable $\mathbf{d}$
Linear Regression	$y_i^* = y_i - (\beta_1 d_1 + \dots + \beta_D d_D)$	β : coefficients of the linear regression using y_i as dependent variable, and $\mathbf{d}$ as independent variables

The main difference between both methods is that the linear regression allows for controlling for multiple variables. One should mention that this is only applicable for continuous outcome systems; one needs to ensure that the controlling variables have a certain sample size, and therefore cut-offs and segregation may be required.



Other considerations

Under the Equal Employment Opportunity Commission's Uniform guidelines, under some specific assumptions and circumstances, adverse impact is said to be occurring when the selection rate of one subgroup is less than four-fifths (80%) of the group with the highest selection rate. Though it may not be applicable in most circumstances, this criterion is used as a support flag during the audit to determine to evaluate the impact ratios. This should only be used as a guidance, rather than a diagnostic or statement regarding adverse impact, given that such assessment is beyond the bias audit as per New York City Local Law 144 and DCWP rules.

In line with the EEOC's guidance on their Questions and Answers to Clarify and Provide a Common Interpretation of the Uniform Guidelines on Employee Selection Procedures, specifically referencing Question 16, a small sample size is defined as a group representing less than 2% of the total labor force in the relevant labor area or 2% of the applicant pool. Under the adopted rules, auditors are permitted to exclude a category that represents less than 2% of the dataset being used for the bias audit from impact ratio calculations providing the summary of results includes a justification for doing so and the number of applicants and scoring rate or selection rate of the excluded category. Thus, if a group with a small sample size has the highest rate or score, we do not use this as the denominator for the impact ratio metric and instead use the group with the highest score/rate that has a sufficient sample size. Therefore, the impact ratio for the underrepresented group would be greater than 1. Statistics and statistical claims are unreliable under insufficient sample size regimes.

1.4. Reporting

Using the proprietary Holistic AI Auditing Platform, two collaterals are generated: Bias Audit Report and Summary of Results. The key elements composing both and disclosure type are summarized in Table 8.

Table 8. Audit outcomes, disclosure type and information

Audit Outcome	Disclosure type	Information
Bias Audit Report	Internal. Can be made available or shared to a client	-Statement by Service Organization Management - Executive Summary - Methodology - Descriptions - Audit Results - Recommendations
Summary of Results	External. Need to be made available on the employment section of their website, job board, and any other channel used by candidates or employees, with an active hyperlink.	- Date of audit - System description - Key information used - Data used to conducted - Impact ratios and others

The Bias Audit report (this document) is generated outlining the details of the System being audited, the methodology used to conduct the audit, key findings, and recommendations. Findings are presented in terms of impact ratios, calculated using the above metrics. Holistic AI's auditing team then performs a review of the outcome of the audit against Local Law 144 standards, using the four-fifths threshold as a guide to determine whether exceptions should be further analyzed. Any analyses based on small sample sizes are indicated with an asterisk.



In addition, a Summary of Results of the bias audit publicly available in the form of an interactive dashboard with findings at the standalone and intersectional level available for both system-level analysis (i.e., all data provided) and company-level analysis (i.e., results by companies represented in the dataset) and is publicly accessible via a URL given to the Service Organization that can be linked on their website.

1.5. Ongoing Monitoring

Given that AI and other automated systems regularly change and are updated as additional data and insights become available, periodic monitoring is often not sufficient to ensure the safety of the system. As such, in addition to the annual bias audits required by NYC LL144, Holistic AI can also audit the system as needed following any major updates or modifications or following the implementation of relevant mitigation procedures to examine their effectiveness at reducing bias or the risk of other relevant verticals. In the case of additional audits, the audit report and summary of results are reproduced to reflect the latest changes.

In line with the notification requirements of Local Law 144, where candidates and employees must be informed of the use of the tool at least 10 business days before its use, in addition to information about the System data retention policy and how an alternative selection procedure or accommodation can be requested, Holistic AI provides guidance documentation with sample notices.



2. Service Organization Details

2.1. Company Profile Description

Maki People (Maki) is a Paris-based HR technology company founded in 2021. The company develops an AI-powered talent assessment and hiring platform designed to help organisations screen, evaluate, and select candidates at scale.

The company's core product is an end-to-end hiring platform that deploys autonomous AI agents to manage key stages of the recruitment process, including candidate screening, skills testing, and interviewing. These agents conduct voice and web-based assessments covering over 300 skill dimensions, ranging from cognitive ability and language proficiency to behavioural traits and role-specific technical skills. The platform is designed to integrate with existing Applicant Tracking Systems, sitting as an intelligence layer on top of an organisation's existing recruitment infrastructure.

The platform uses speech-to-text engines and large language models to evaluate candidates in a conversational format, generating standardised, psychometrically validated scores intended to support more consistent and defensible hiring decisions.

2.2. System Audited

Maki People's Shiro – Shiro is an AI-assisted decision-support assessment platform that evaluates 19 soft skills using two standardized open-ended questions per skill (broad and scenario-based), applying light preprocessing, Behaviourally Anchored Rating Scales, and a large language model to assign dimension-level scores which are aggregated into evidence-based scorecards with quantitative results and qualitative explanations to support human hiring decisions, and it does not autonomously make hiring decisions.

2.3. Design and Development Team

The following table includes the name, contact details, and role of the individuals that contributed to the design and/or development of the System under audit.

Table 9. Design and development team

Name	Contact Details	Role
Dr Aiden Loe	aiden.loe@makipeople.com	Head of Science
Mr Benjamin Chino	chino@makipeople.com	Chief of Product
-	-	Science x Tech Team

2.4. NYC Local Law 144 Qualification

Automated decision systems that use a computational process derived from machine learning, statistical modeling, data analytics, or artificial intelligence that produce a simplified output (a score, classification, or recommendation) qualify for an audit if they are used to substantially assist or replace human decision making about employment decisions.

**Table 10.** NYC Local Law 144 qualification criteria

Category	Qualifications If Used to Substantially Assist or Replace Human Decision Making about Employment Decisions
Company	Employer
	Employment agency
	Used by Employers or Employment Agency
Other	Career Sites
	Candidate Relationship Management
	Applicant Tracking System
	Search and Matching Services
	Assessments
	Talent Mobility / Internal Hiring

The legislation clarifies that the group of systems that qualify for an audit are those that generate a prediction of a candidate's fit or likelihood of success, or that generate a classification based on skill sets or aptitude. A computer must at least in part identify the inputs, the relative importance placed on those inputs, and other parameters for the models in order to improve the accuracy of the prediction or classification. Based on the information provided, Shiro meets at least one of the technical requirements of the legislation; however, it must be used to substantially assist or replace discretionary decision making, which is dependent on the client's use of the tool to be considered an automated employment decision tool (AEDT).

Table 11. NYC Local Law 144 qualification assessment

Category	System Qualification If Used to Substantially Assist or Replace Human Decision Making about Employment Decisions
Method	Uses machine learning
	Uses statistical modelling
	Uses data analytics
	Uses artificial intelligence
Output	Used for scoring
	Used for ranking
	Predicts fit or likelihood of success
Usage	Used in hiring decisions
	Used in promotion decisions



3. System Description for Shiro

Shiro is an AI-assisted assessment platform designed to evaluate candidate soft skills through structured, open-ended written responses. The system generates standardised, evidence-based scorecards to support human-led hiring decisions. Shiro operates strictly as a decision-support tool and does not autonomously make hiring decisions.

Shiro operates through a structured, multi-stage processing pipeline that ensures consistency, traceability, and reproducibility of scoring outcomes. Candidate responses are first collected through standardised open-ended questions, with each response linked to a specific skill, behavioural dimension, and question type (broad or scenario-based). These responses undergo light preprocessing to standardise formatting without altering semantic content. The processed responses are then evaluated against predefined Behaviourally Anchored Rating Scales (BARS). A large language model assesses each response by comparing it against dimension-specific behavioural criteria and evidence thresholds. Scores are assigned independently at the dimension level using a fixed scoring rubric. Following scoring, dimension-level scores are aggregated using predefined weights to produce overall skill scores. These scores are then compiled into structured scorecards, which include quantitative results alongside qualitative explanations. This pipeline ensures that all candidates are assessed using the same evaluation logic and transformation steps.

The assessment framework evaluates 19 soft skills, each measured using two open-ended questions. The first question is designed to elicit a broad, contextual response, while the second probes more deeply into a specific scenario. This dual-question approach is intended to capture both the range and depth of a candidate's behaviour, reducing reliance on self-perception and increasing the likelihood of observing concrete behavioural evidence.



4. Audit Results

4.1. *Data Extraction and Quality Assurance*

Data Provided. Information provided by Maki People:

- Data type: test
- Region: N/A
- Pre-cleaning sample size: 3920
- Number of missing values: 0
- Sample size after removing missing values: 3920

Additional Data Details. Synthetic interview transcripts were generated using OpenAI's GPT-5-mini model (via Azure) across a controlled factorial design. The dataset spanned 40 distinct job roles drawn from U.S. Bureau of Labor Statistics occupational categories, ranging from entry-level positions (e.g., Janitor, Cashier, Bartender) to professional and managerial roles (e.g., Financial Manager, Software Developer, Corporate Lawyer). Each job was paired with a detailed job description to ground the AI-generated responses in role-relevant context.

Response quality was controlled using 7 calibrated quality levels — Very Poor, Poor, Below Average, Average, Above Average, Good, and Excellent — each defined by specific persona instructions and word-count targets ranging from 70 to 125 words per answer. These levels constrained reasoning depth, example specificity, and degree of self-reflection to produce a realistic distribution of candidate performance.

Demographic identity was embedded into each transcript through 4 placeholder tags that were post-processed into natural self-identification phrases aligned with 14 intersectional demographic groups — 7 EEO-1 race/ethnicity categories (Hispanic or Latino, White, Black or African American, Native Hawaiian or Pacific Islander, Asian, Native American or Alaska Native, and Two or More Races) crossed with 2 sex categories (Male and Female). To test the tool's sensitivity to demographic signals, at least one intentional misspelling of the demographic phrasing was included per transcript.

The raw inputs, which consisted of two question answers pairs (question and follow-up question) per dimension, per candidate profile (Job x Quality x Gender x Ethnicity combinations) were scored individually (74,480 questions) and aggregated to form a fully scored simulated interview.

In total, the final dataset comprised 3,920 unique simulated interviews — 280 per intersectional demographic group and 560 per race/ethnicity group — scored across the full matrix of 40 job roles, 7 quality levels, and 14 demographic groups. These transcripts were submitted to Maki's Mochi scoring system for evaluation, and the resulting scores were aggregated for statistical bias analysis by demographic group.

Data cleaning. The steps performed are as follows:

- The original dataset contained 3,920 records.
- Ethnicity values were mapped as follows: the original value "Native American or Alaska Native" was mapped to "American Indian or Alaska Native"; the original value "Native Hawaiian or Pacific Islander" was mapped to "Native Hawaiian or Other Pacific Islander".
- After data cleaning, 3,920 records remained for analysis (100.0% of original data).
- Missing data: N/A
- Post-cleaning sample size: 3920



Norming. To produce system-level (overall) and company-level impact ratios, the following methodology was applied:

- System type: Continuous outcome
- Range of the outcome: [3.421, 13.526]
- Encoding: categorical variables were one-hot encoded, and when they are binary, the first class within a category is dropped.
- Outliers: presence of outliers were investigated, and model-based approaches were employed in case it needed mitigation.

EEOC small sample size flag – unreliable statistic. As mentioned in the methodology, we define a small sample as the group representing less than 2% of the individuals.

4.2. Exploratory data analysis

File hash and last date modified.

Table 12. File hash and last date modified

File Type	File Name	md5 checksum	Last Modified
Clean data	set_1234_aggregated_avg.csv	b998e8189cec4a8ba7bf042ab7097355	March 23, 2026 at 5:01:07 PM

Protected attributes.

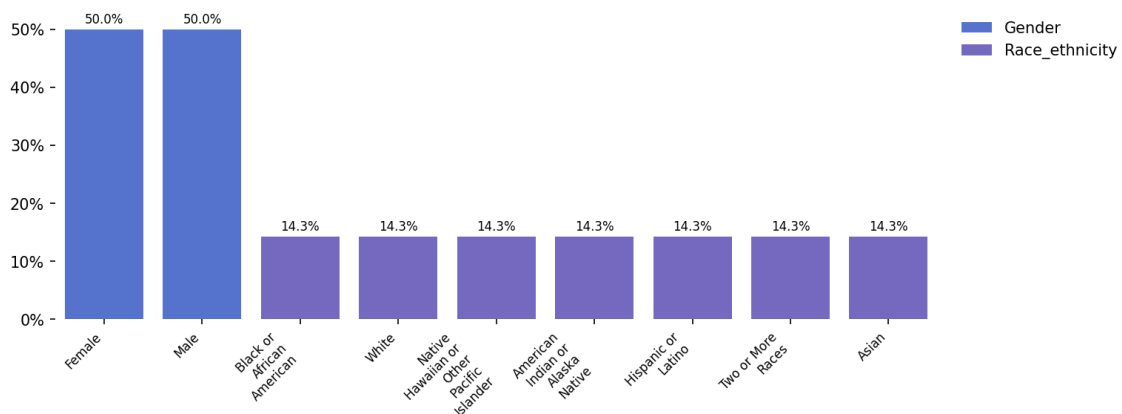


Figure 2. Number of individuals in each group

Score statistics

Table 13. Scoring range statistics

Stat	Value
min	3.42



max	13.53
median	7.16
std	3.12
skew	0.29

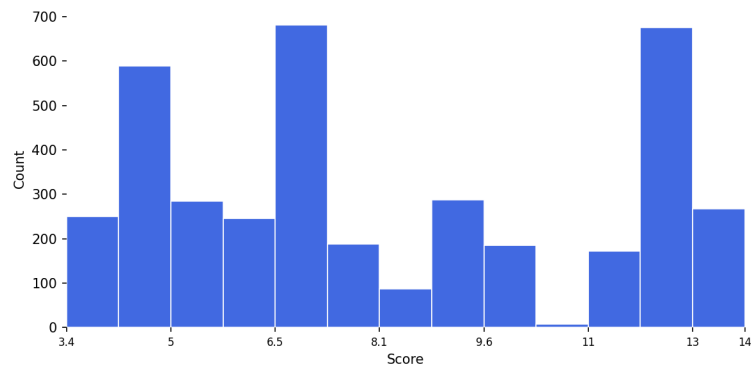


Figure 3. Aggregated score distribution

4.3. Impact Ratios Results – Standalone

In this section, we present results based on the impact metrics outlined in section 1 for Gender, Ethnicity.

- **Summary of the findings:** Standalone analysis was conducted across gender and race/ethnicity groups using the scoring rate methodology for continuous outcome systems. For gender, the scoring rate was 0.50 for female candidates and 0.50 for male candidates, resulting in an impact ratio of 0.99. Across race and ethnicity groups, scoring rates ranged between 0.45 and 0.53, with impact ratios ranging between 0.85 and 1.00 relative to the highest scoring group. The impact ratio for White candidates was 0.85, which meets the four-fifths (0.80) threshold. All standalone impact ratios exceeded the four-fifths reference threshold commonly used in employment bias audits, indicating no material disparities in scoring outcomes across the demographic groups evaluated.
- **Low number groups:** Certain demographic categories may have relatively smaller sample sizes depending on the deployment context. In accordance with EEOC guidance and NYC Department of Consumer and Worker Protection (DCWP) rules, demographic groups representing less than 2% of the dataset may be excluded from impact ratio calculations where statistical reliability cannot be ensured. Where applicable, these exclusions are documented and clearly indicated in the audit results.

Table 14. Standalone impact ratios - Gender

Gender	Scoring Rate	Impact Ratio	Sample Size
Female	50.0%	1.00	1960
Male	49.7%	0.99	1960

–Group represents less than 2% of individuals, impact ratio can be excluded from analysis.

Table 15. Standalone impact ratios - Ethnicity

Ethnicity	Scoring Rate	Impact Ratio	Sample Size
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Black or African American	51.1%	0.97	560
White	44.8%	0.85	560
Native Hawaiian or Other Pacific Islander	52.3%	0.99	560
American Indian or Alaska Native	52.7%	1.00	560
Hispanic or Latino	50.7%	0.96	560
Two or More Races	49.3%	0.94	560
Asian	47.9%	0.91	560

–Group represents less than 2% of individuals, impact ratio can be excluded from analysis.

4.4. Impact Ratios Results – Intersectional

In this section, we present the impact ratios based on intersectional groups. In other words, we tested for adverse impact at the intersection of two demographic groups.

- **Summary of the findings:** Intersectional analysis was conducted across combined gender and race ethnicity groups to evaluate potential disparities at the intersection of protected characteristics. Scoring rates across intersectional groups ranged between 0.44 for males who are White and 0.53 for females who are American Indian or Alaska Native, with impact ratios ranging between 0.82 and 1.00 relative to the highest scoring group. All intersectional impact ratios were above the four-fifths (0.80) threshold used as a reference guideline in bias audits, suggesting consistent scoring outcomes across the intersectional demographic groups evaluated.
- **Low number groups:** Some intersectional demographic groups may represent smaller portions of the dataset, which can result in greater statistical variability when estimating impact ratios. In line with regulatory guidance, intersectional groups representing less than 2% of the dataset may be excluded from formal impact ratio calculations when statistical reliability cannot be ensured. Any such exclusions are clearly documented in the audit results.

Table 16. Intersectional impact ratios - Gender/Ethnicity

Gender	Ethnicity	Scoring Rate	Impact Ratio	Sample Size
Female	Black or African American	51.4%	0.97	280
Female	Two or More Races	48.6%	0.92	280
Female	Hispanic or Latino	51.1%	0.97	280
Male	Two or More Races	50.0%	0.95	280
Male	American Indian or Alaska Native	52.5%	0.99	280
Female	White	46.1%	0.87	280
Male	Asian	48.2%	0.91	280
Male	Black or African American	50.7%	0.96	280
Female	Asian	47.5%	0.90	280
Male	White	43.6%	0.82	280
Female	American Indian or Alaska Native	52.9%	1.00	280
Male	Native Hawaiian or Other Pacific Islander	52.5%	0.99	280
Male	Hispanic or Latino	50.4%	0.95	280
Female	Native Hawaiian or Other Pacific Islander	52.1%	0.99	280

–Group represents less than 2% of individuals, impact ratio can be excluded from analysis.



5. Recommendations

Annual audit as required by NYC LL144 with the next date set at March 23, 2027.

Although no impact ratios fell below the 0.80 threshold indicating adverse impact, some groups such as White individuals and related intersectional subgroups approached this boundary, suggesting areas for continued vigilance. The company should maintain robust internal monitoring processes to routinely assess scoring outcomes across all protected groups to detect any emergent disparities promptly. Implement periodic fairness reviews integrated into model performance evaluations to ensure ongoing compliance with NYC Local Law 144. It is important to document all monitoring activities and remediation efforts thoroughly to meet regulatory expectations. Additionally, consider implementing bias mitigation strategies proactively, such as algorithmic adjustments or fairness-aware training techniques, especially targeting subpopulations with impact ratios nearing concern levels. Engage diverse stakeholders in reviewing outcomes and refining decision criteria to uphold equitable treatment. By establishing a continuous fairness governance framework and committing to transparent reporting, the organization can sustain equitable results over time and reduce risk of inadvertent bias in future deployments.

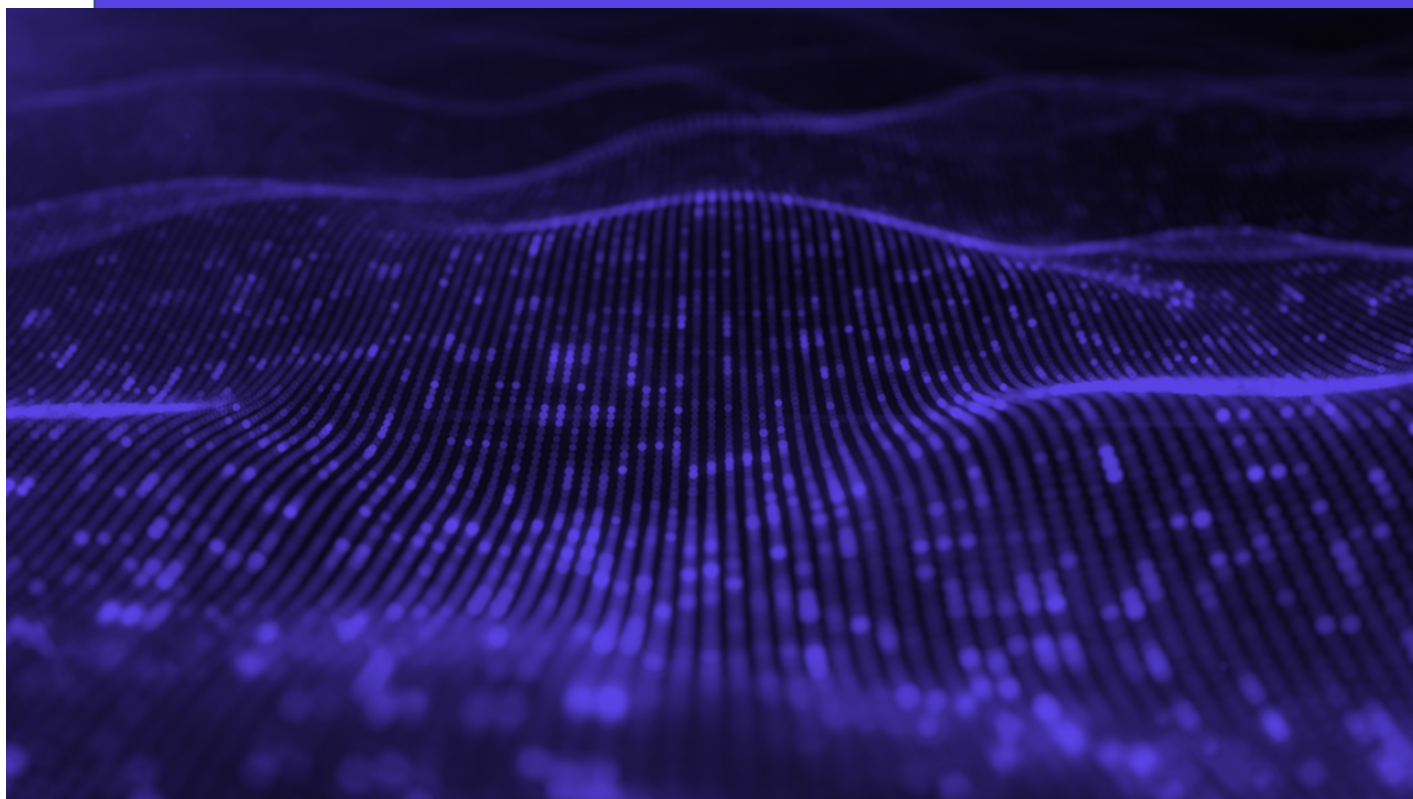


6. Deployment Data Provided

Deployment Information. The table below shows the total number of datapoints for each protected characteristic provided and each deployment using Maki People's platform.

Table 17. Number of samples per deployments and protected attribute

Deployments	Gender	Ethnicity
Sales Manager	98	98
Security Guard	98	98
Plumber, Pipefitter, and	98	98
Welders, Cutters,	98	98
Laborer and Freight,	98	98
Retail Salesperson	98	98
Compliance Officer	98	98
Heavy and Tractor-Trailer	98	98
Licensed Practical and	98	98
Project Management	98	98
Cook, Restaurant	98	98
Software Developer	98	98
Customer Service	98	98
Accountant/Auditor	98	98
Medical Assistant	98	98
Financial Manager	98	98
Bartender	98	98
Home Health and Personal	98	98
Police and Sheriff's Patrol	98	98
Human Resources	98	98
Teaching Assistant	98	98
Paralegal and Legal	98	98
Carpenter	98	98
Corporate Lawyer	98	98
General Office Clerk	98	98
Cashier	98	98
Registered Nurse	98	98
Computer and	98	98
Elementary School	98	98
Childcare Worker	98	98
Insurance Sales Agent	98	98
Operating Engineer and	98	98
Electrician	98	98
General and Operations	98	98
Landscaping and	98	98
Hairdresser/Cosmetologist	98	98
Management Analyst	98	98
Pharmacy Technician	98	98
Automotive Service	98	98
Janitor/Cleaner	98	98



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nyc-bias-audit-main-report-shiro-maki

Final Audit Report

2026-03-27

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